ISSUES REPORTING OUTLET (IRO)

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1. Project Description

I developed a web-based prototype called Issues Reporting Outlet (IRO) to help students and residents around AAUA report community problems such as bad roads, electricity issues, water shortages, drainage problems, and waste disposal.

2. Questionnaire

I prepared a questionnaire using Google Forms to understand the common problems students experience in their communities and how they currently report them.

The questions focused on:

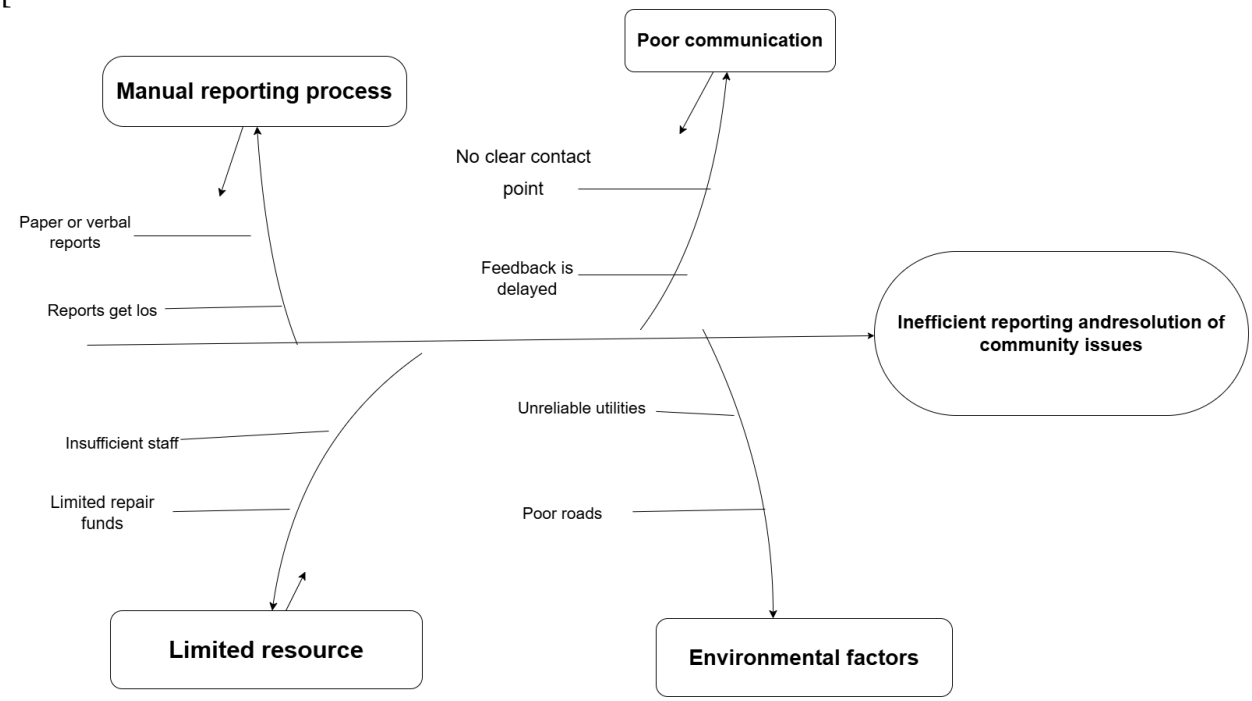
- Common community problems.
- How often these problems occur.
- How people currently report issues.
- Challenges people face when reporting problems.
- Whether people would use an online reporting platform.
- Features users would like to see in the system.

The responses collected will help identify the most common issues and guide the design of the system. Actual response counts and percentages should be added from the Google Forms results.

3. Problem Analysis

I used an Ishikawa (Fishbone) Diagram to analyse the possible causes of community problems and reporting difficulties. The diagram helps organise the causes and show how they contribute to the identified problems.

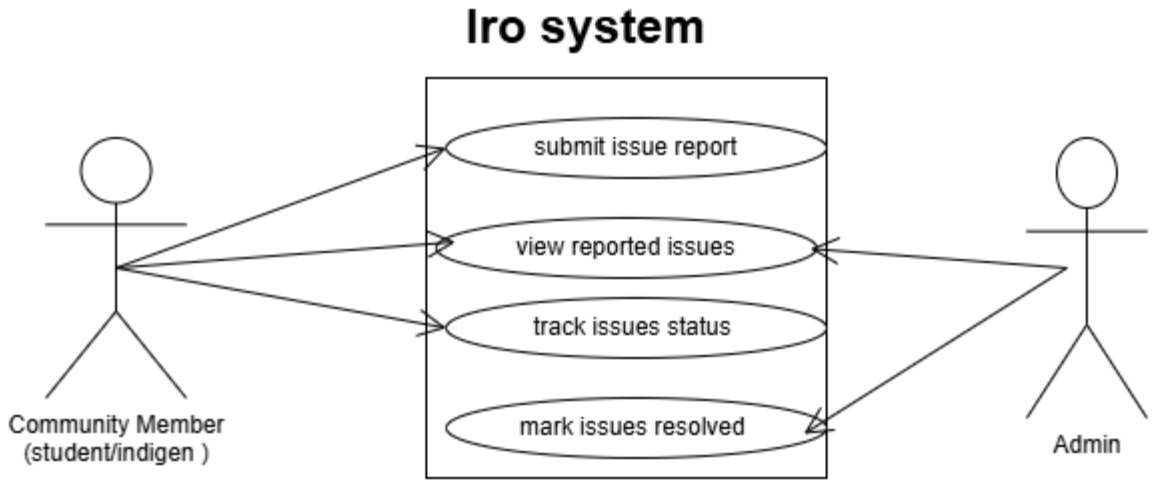
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Fishbone Diagram here]

4. System Design

I created a use-case diagram to show how users interact with the IRO system. Users can access the homepage, select an issue category, enter details about a problem, submit a report, and search for a report using its reference number.



use case diagram of the iro system

[
Use-Case Diagram here]

5. Object Listing

The main objects and data used in the system include:

1. User: Represents the person submitting a report.
2. Issue Report: Stores the details of a reported problem.
3. Issue Category: Identifies the type of problem, such as electricity, water, or roads.
4. Reference Number: Helps users identify and retrieve their submitted reports.
5. Report Form: Collects information about the issue.
6. Tracking Section: Allows users to search for a report using its reference number.

6. Tools and Technologies Used

- HTML: Used to structure the web pages.
- CSS: Used to style the interface and make it user-friendly.
- JavaScript: Used to handle form interactions, generate reference numbers, and retrieve saved reports.
- Browser Local Storage: Used to save reports in the current browser.
- GitHub: Used to store the project's source code.

7. Implementation

I designed the homepage with a green-themed interface and created a form for reporting community issues. The system allows users to enter issue details, submit reports, receive reference numbers, and search for saved reports using those numbers.

The current version is a frontend prototype. Reports are saved in the browser's local storage, meaning they are not yet available to other users on different devices.

8. Testing

I checked the main features of the prototype, including page display, report form interaction, reference number generation, and report searching. Any feature that has not been tested successfully should be checked before marking it as completed.

9. Limitations and Future Improvements

The current prototype does not have a shared database, server-side processing, or a fully functional administrator dashboard. In the future, I would connect it to a database, add user authentication, and provide an administrator interface for managing and updating reported issues.

10. Conclusion

I developed the IRO prototype to provide a simple way for students and residents around AAUA to report community problems. The project helped me apply HTML, CSS, and JavaScript to a practical problem and demonstrate how technology can improve the organisation of community issue reports.